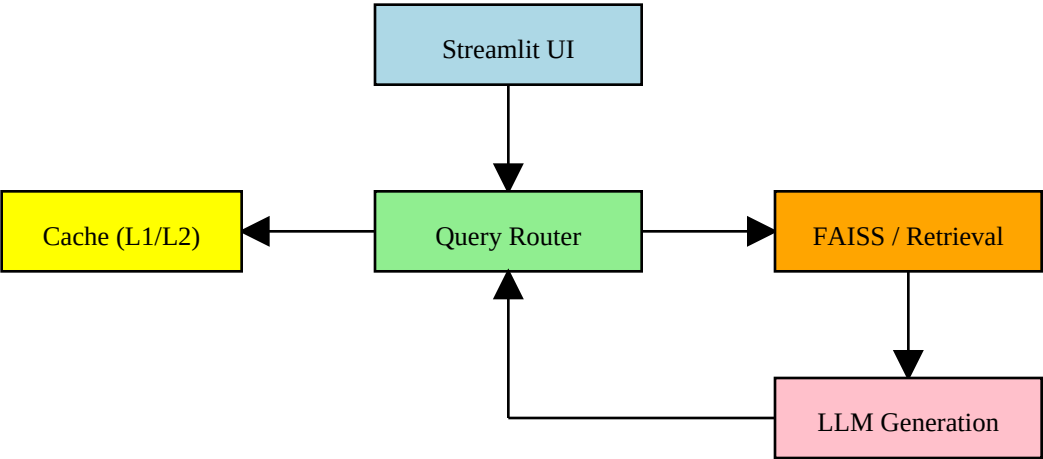


Automotive QA RAG System - Working & Architecture

System Overview

The Automotive QA system is an advanced Retrieval-Augmented Generation (RAG) platform designed to process automotive manuals, queries, and QA flows. The architecture uses a modular approach featuring an interactive Streamlit UI, a routing mechanism, two-tier caching (RAM and SQLite), a vector database (FAISS) for efficient document retrieval, and a Language Model (LLM) for context-aware generation.

System Flowchart



System Components

Component	Technology	Role
UI/Frontend	Streamlit	Interactive user dashboard
Vector DB	FAISS (CPU)	Stores chunk embeddings for fast retrieval
Embedder	SentenceTransformers	Generates vector representations of text
Cache	SQLite / RAM	Improves response time by caching results
NLP Processing	spaCy	Extracts entities and parses text

Report Gen	ReportLab / python-docx	Generates exportable Word/PDF reports
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System Performance Visualization

Visualization displaying expected latency for different system operations.

